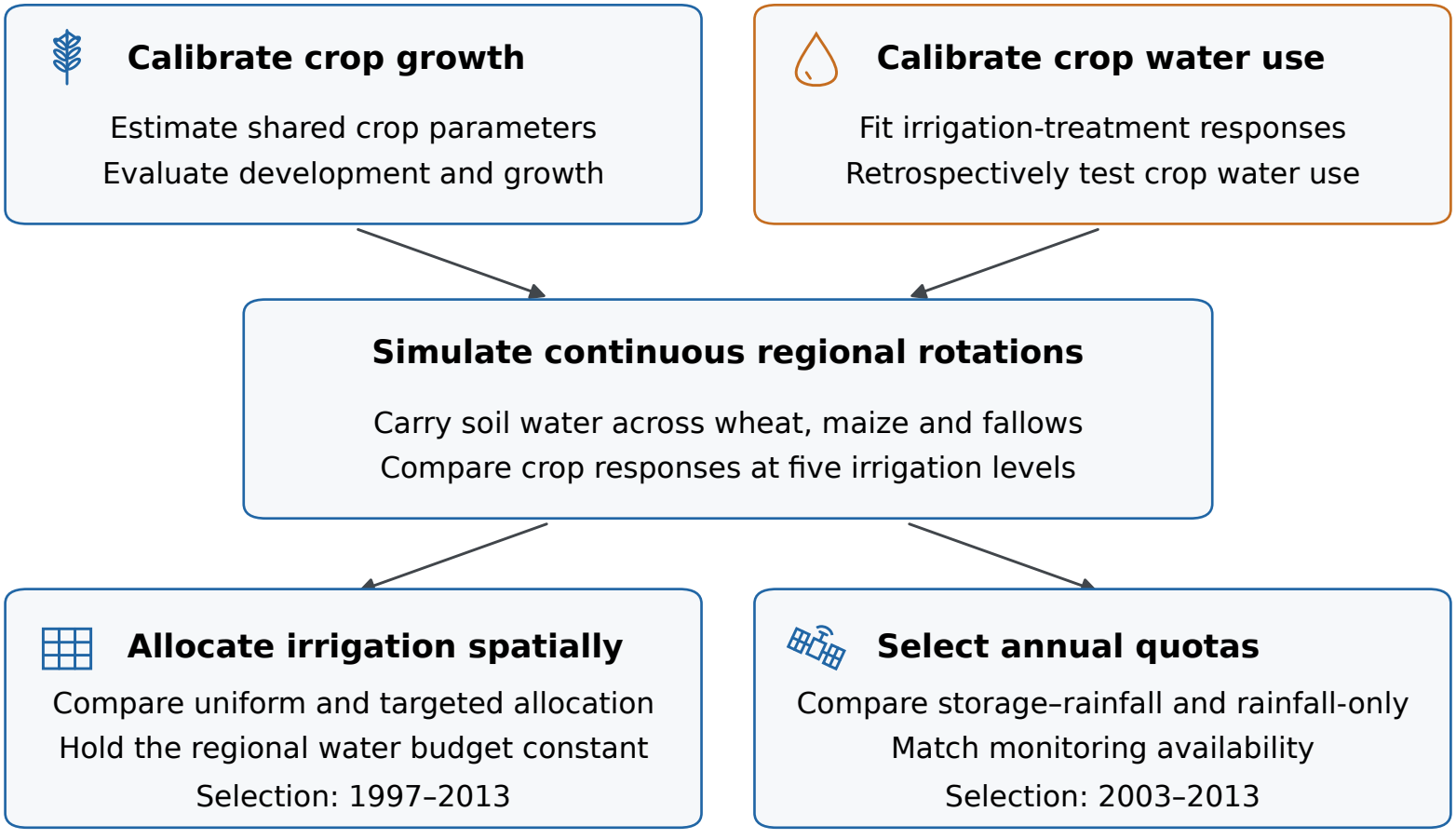
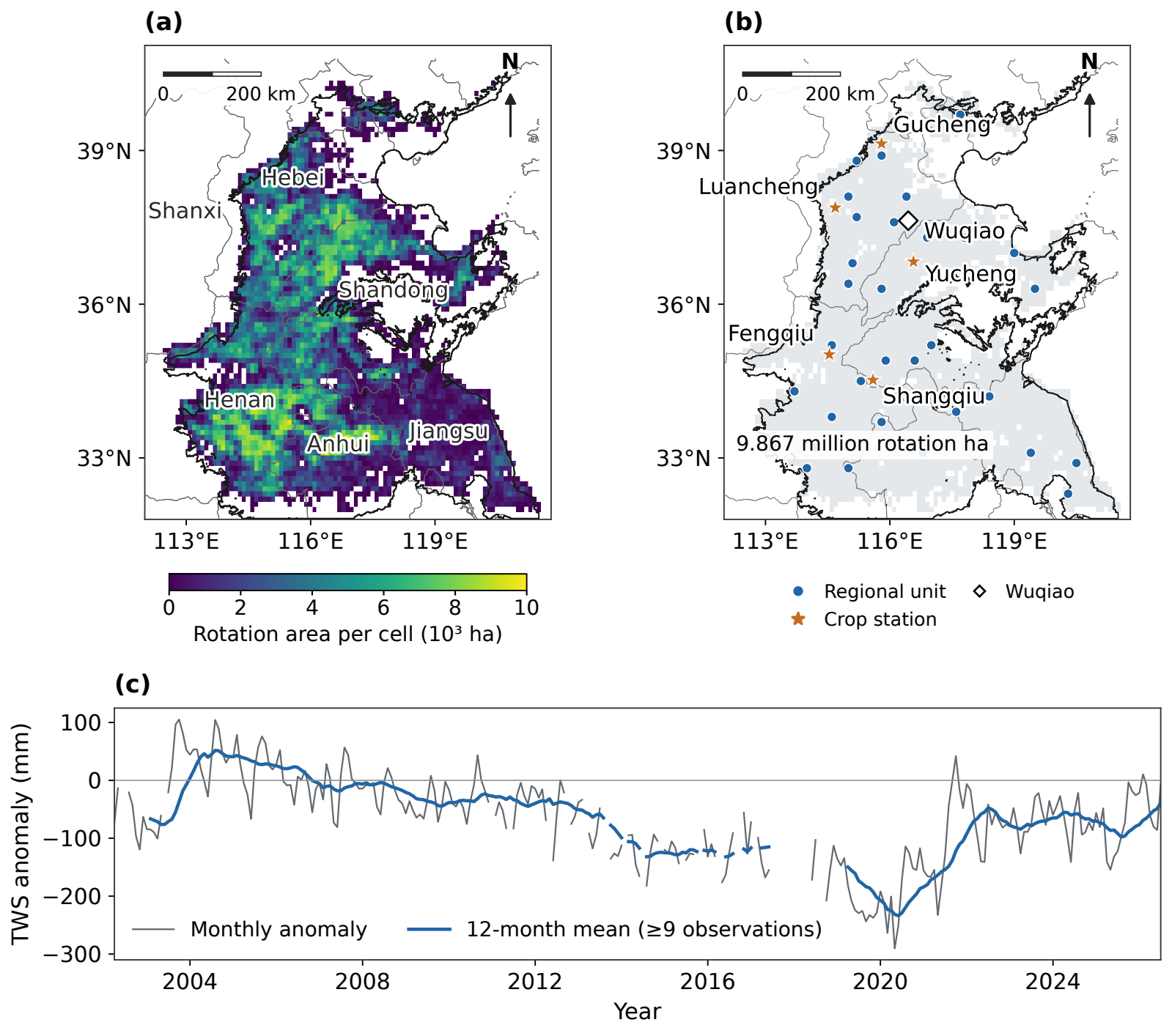
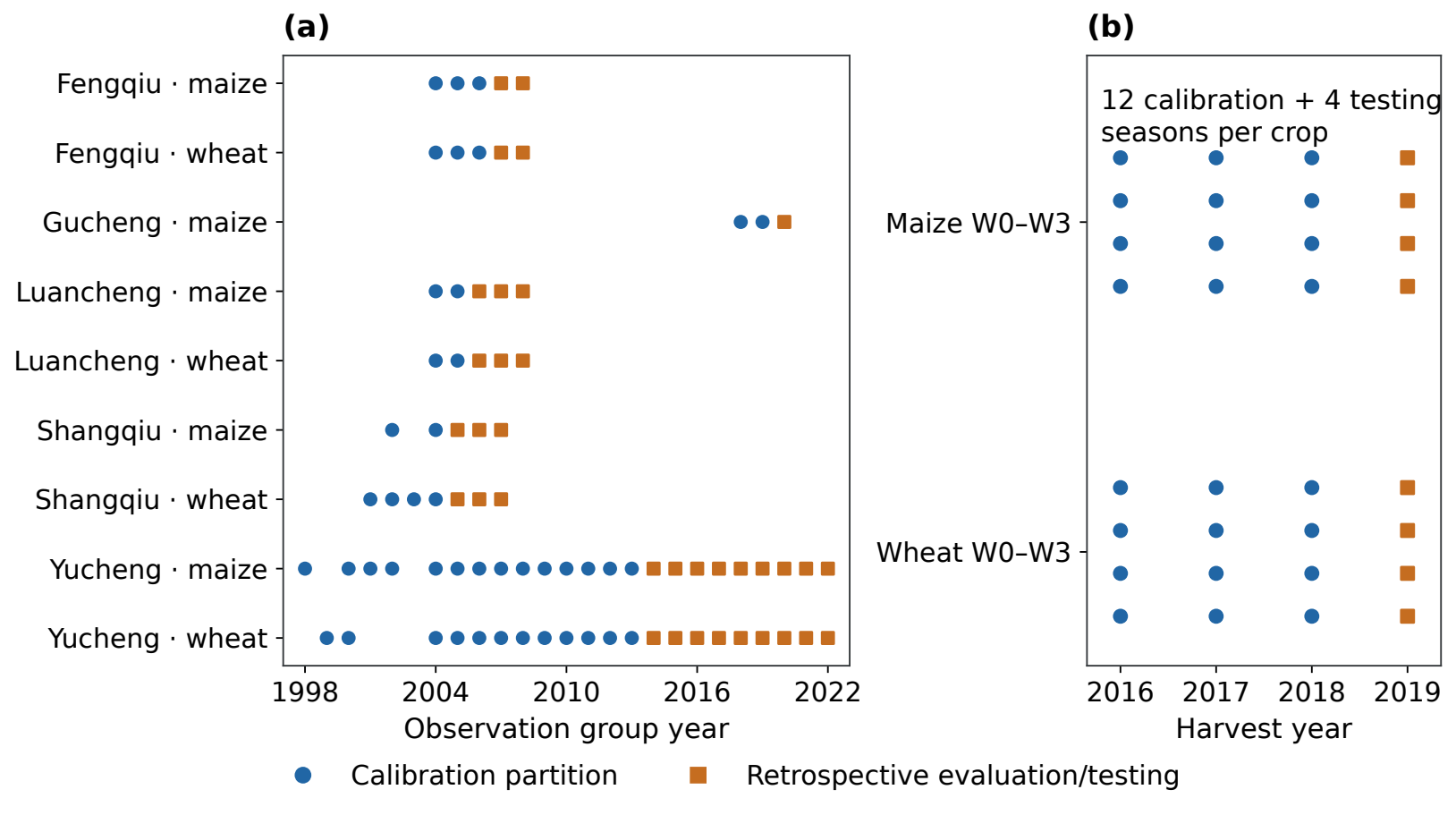
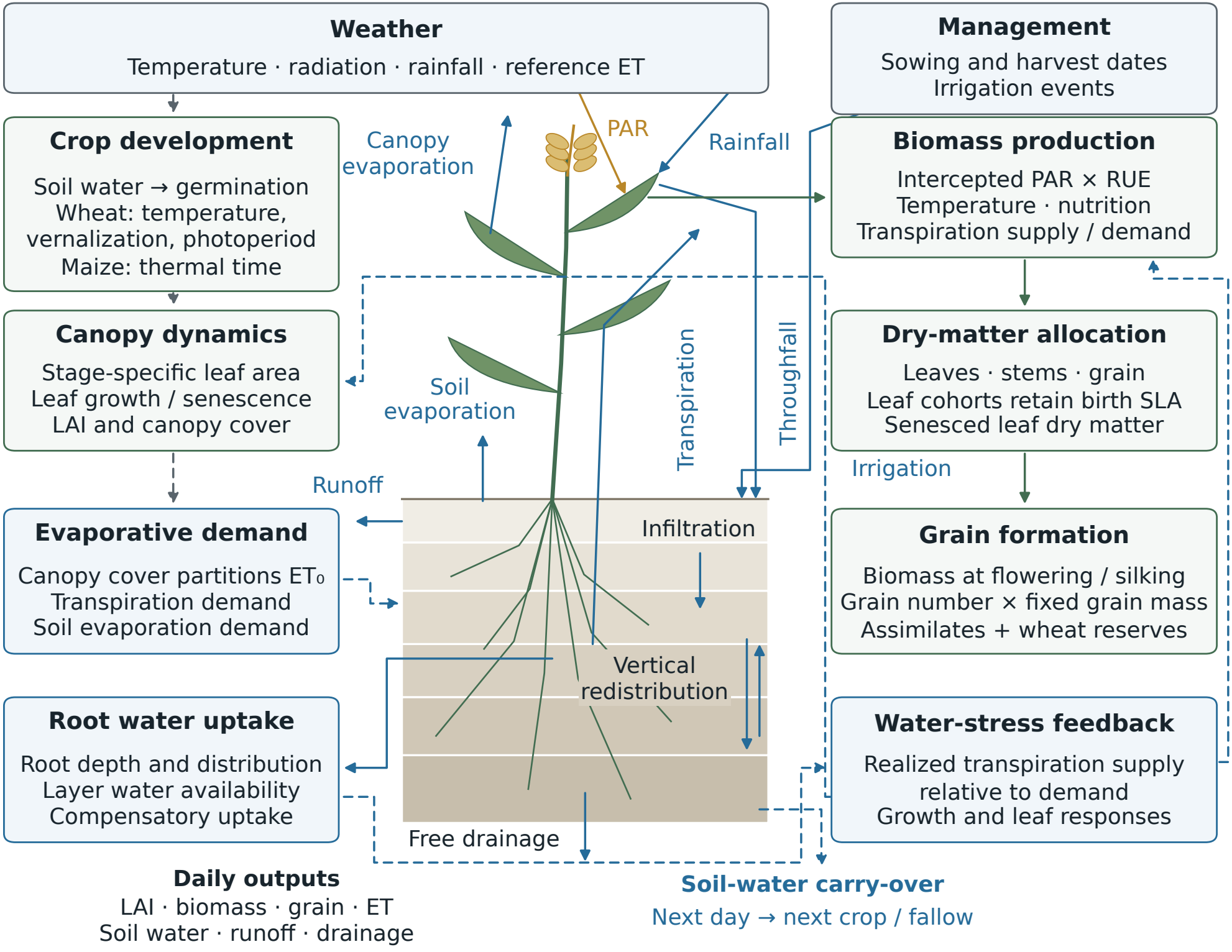
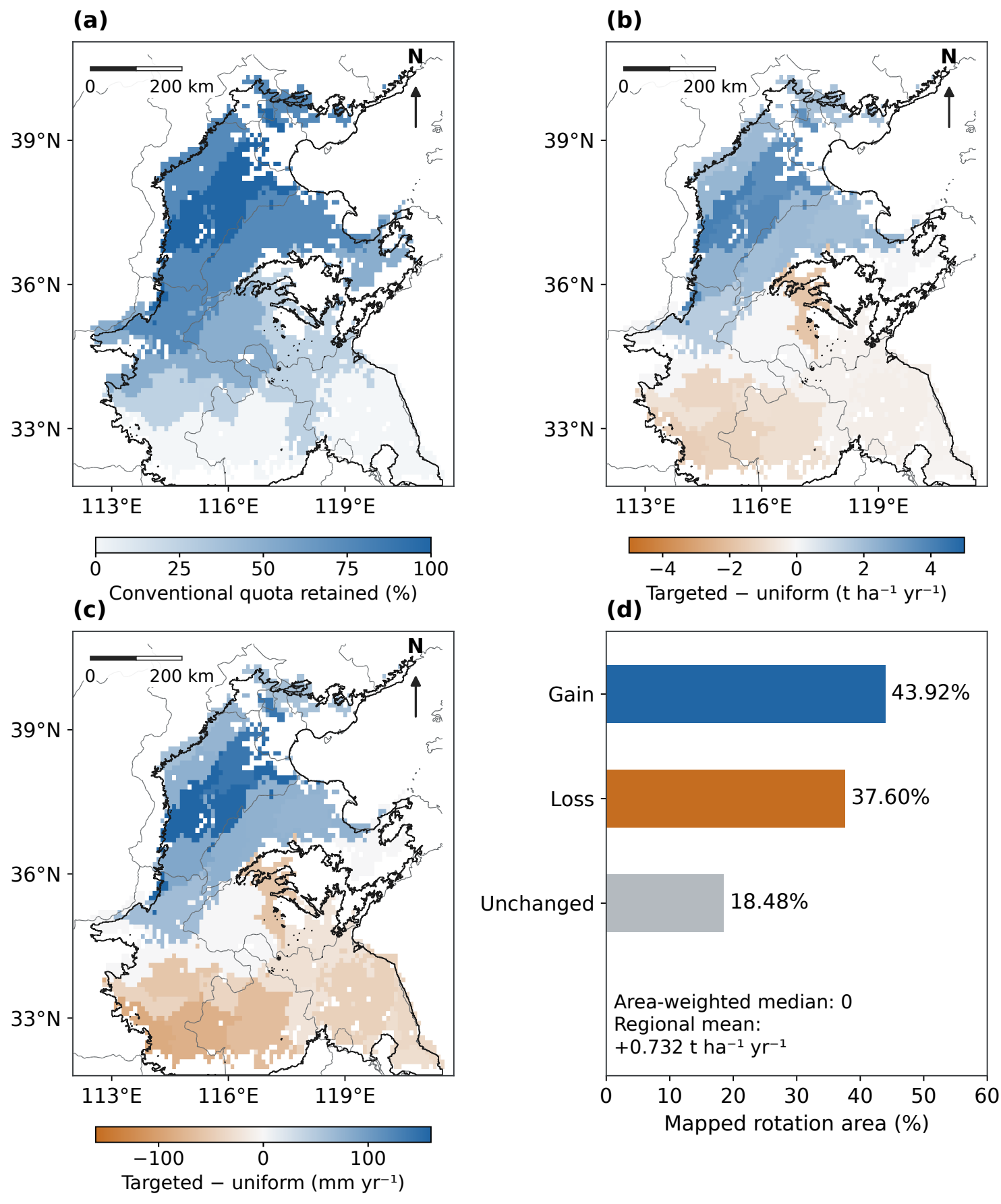
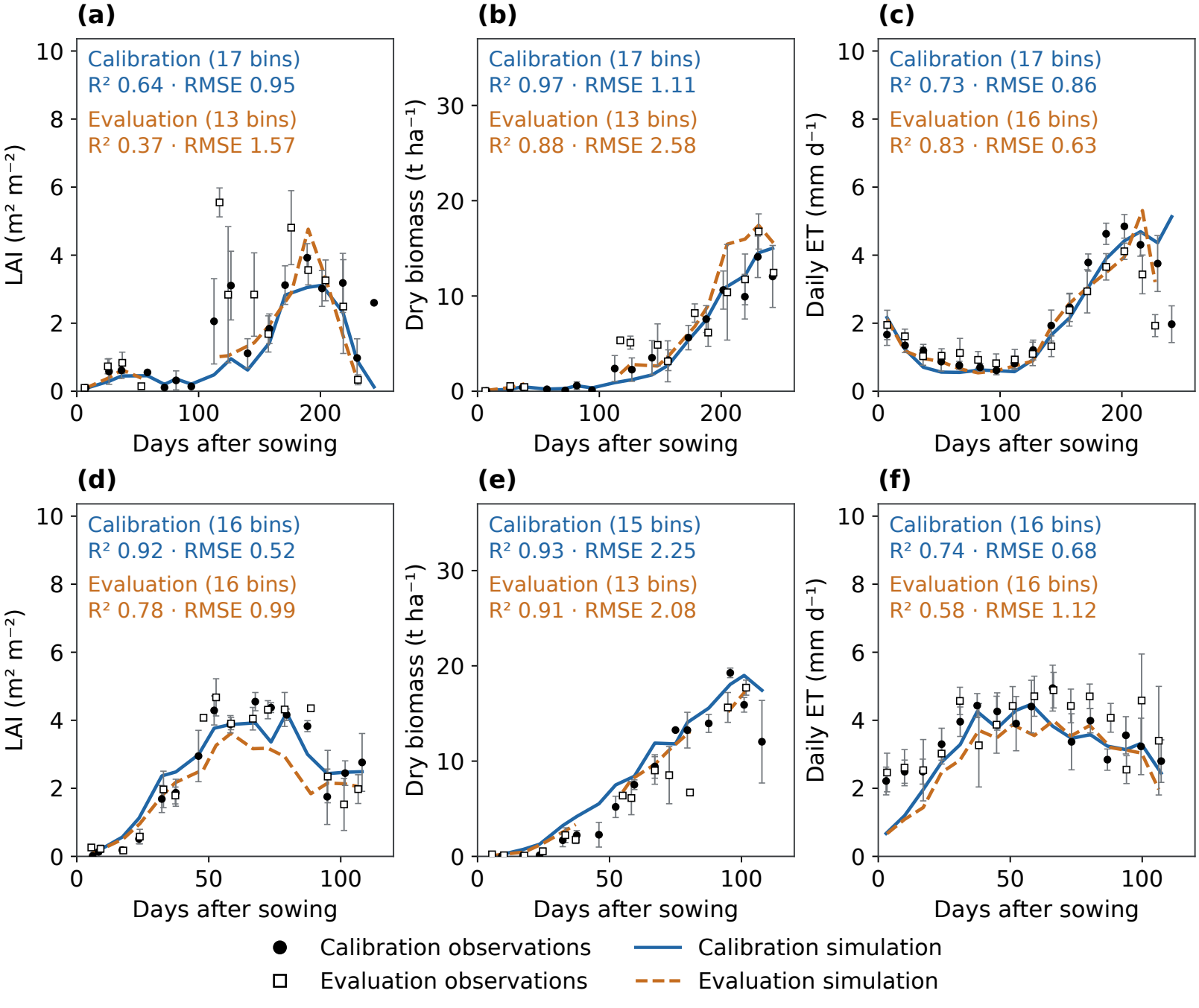

Calibrate crop growth
Estimate shared crop parameters
Evaluate development and growth
Calibrate crop water use
Fit irrigation-treatment responses
Retrospectively test crop water use
Simulate continuous regional rotations
Carry soil water across wheat, maize and fallows
Compare crop responses at five irrigation levels
Allocate irrigation spatially
Compare uniform and targeted allocation
Hold the regional water budget constant
Selection: 1997–2013
Select annual quotas
Compare storage–rainfall and rainfall-only
Match monitoring availability
Selection: 2003–2013
Evaluate management outcomes: 2014–2025
Rotation grain · irrigation · ET · drainage · spatial gains and losses

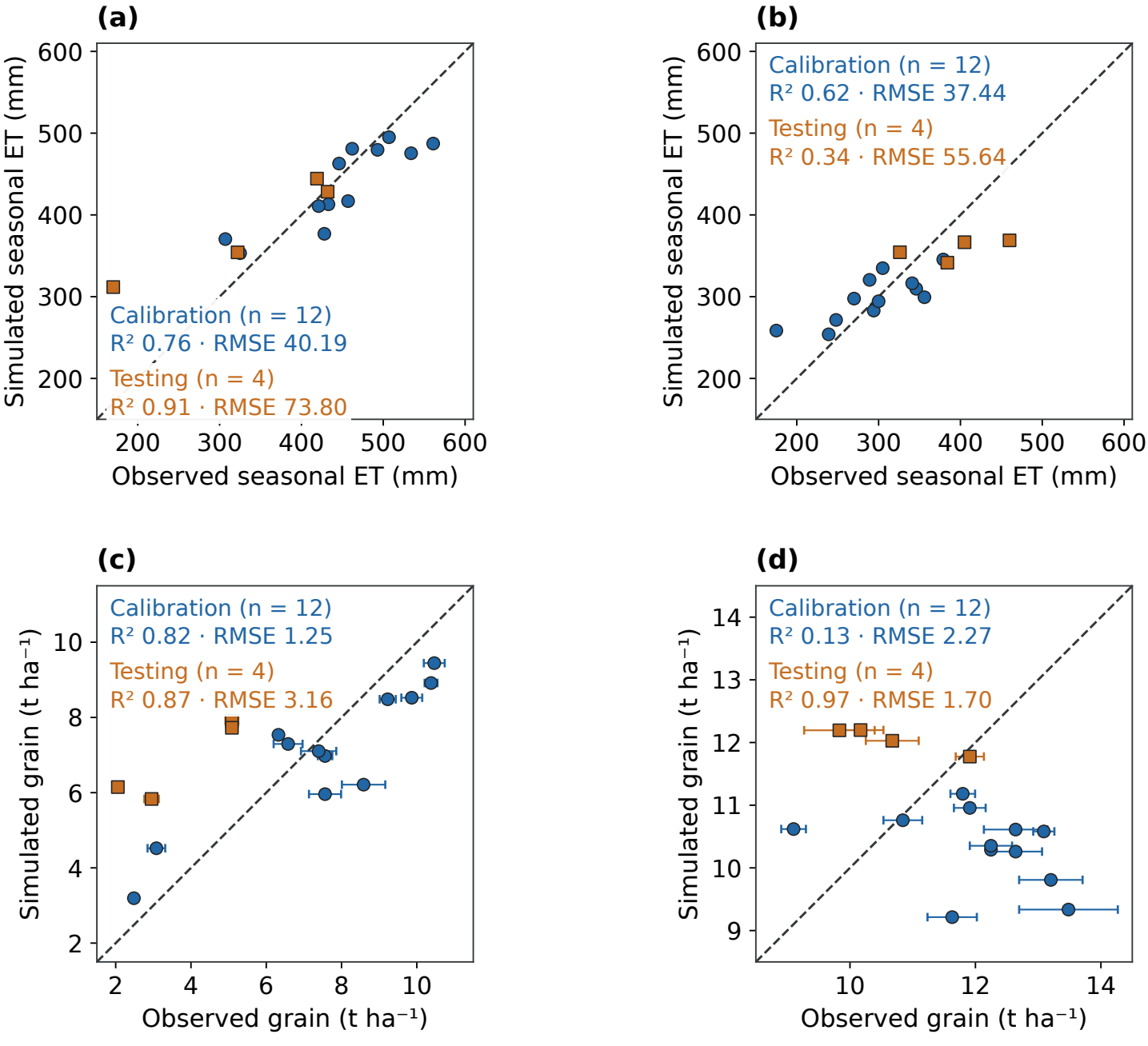

(a)
0
200 km
N
39°N
36°N
33°N
Hebei
Shanxi
Shandong
Henan
Anhui
Jiangsu
113°E
116°E
119°E
0
2
4
6
8
10
Rotation area per cell (10³ ha)
(b)
0
200 km
N
Gucheng
Luancheng
Wuqiao
Yucheng
Fengqiu
Shangqiu
9.867 million rotation ha
39°N
36°N
33°N
113°E
116°E
119°E
Regional unit
Wuqiao
Crop station
(c)
TWS anomaly (mm)
100
0
−100
−200
−300
Monthly anomaly
12-month mean (≥9 observations)
2004
2008
2012
2016
2020
2024
Year

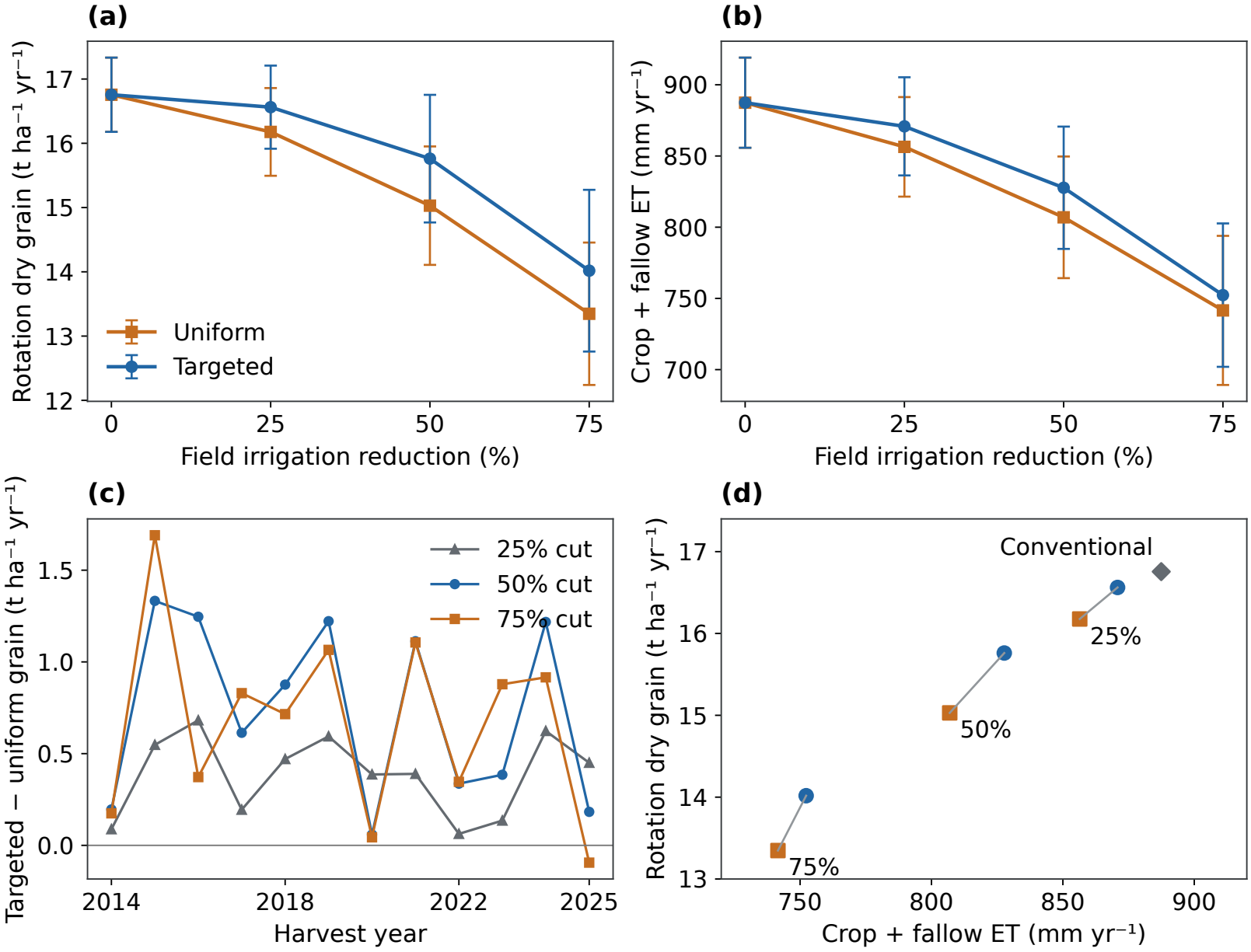

(a)
Fengqiu · maize
Fengqiu · wheat
Gucheng · maize
Luancheng · maize
Luancheng · wheat
Shangqiu · maize
Shangqiu · wheat
Yucheng · maize
Yucheng · wheat
1998
2004
2010
2016
2022
Observation group year
(b)
12 calibration + 4 testing seasons per crop
Maize W0–W3
Wheat W0–W3
2016
2017
2018
2019
Harvest year
Calibration partition
Retrospective evaluation/testing

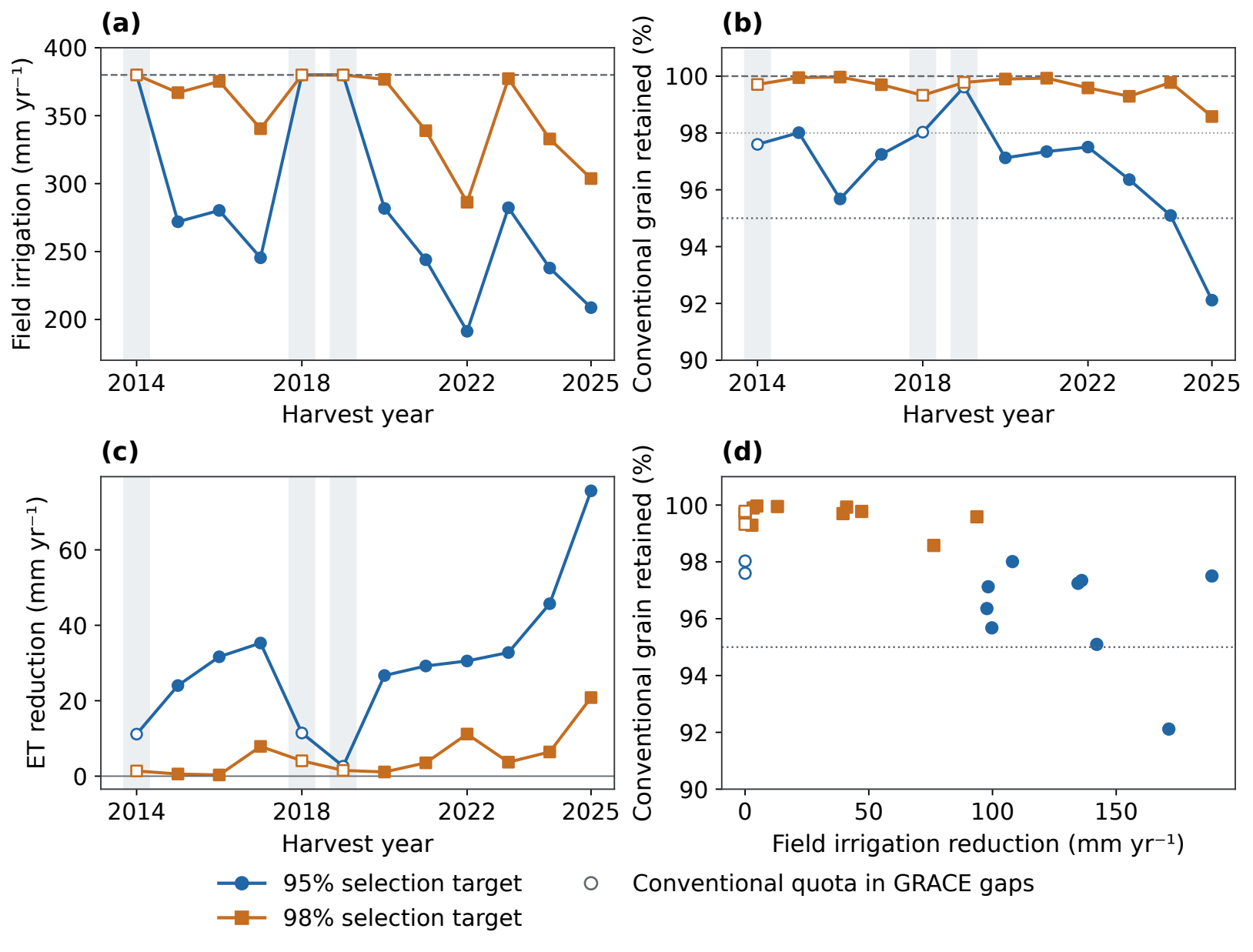

Weather
Temperature · radiation · rainfall · reference ET
Management
Sowing and harvest dates
Irrigation events
Crop development
Soil water → germination
Wheat: temperature, vernalization, photoperiod
Maize: thermal time
Canopy evaporation
PAR
Rainfall
Biomass production
Intercepted PAR × RUE
Temperature · nutrition
Transpiration supply / demand
Canopy dynamics
Stage-specific leaf area
Leaf growth / senescence
LAI and canopy cover
Soil evaporation
Transpiration
Throughfall
Dry-matter allocation
Leaves · stems · grain
Leaf cohorts retain birth SLA
Senesced leaf dry matter
Irrigation
Runoff
Infiltration
Evaporative demand
Canopy cover partitions ET₀
Transpiration demand
Soil evaporation demand
Grain formation
Biomass at flowering / silking
Grain number × fixed grain mass
Assimilates + wheat reserves
Vertical redistribution
Root water uptake
Root depth and distribution
Layer water availability
Compensatory uptake
Water-stress feedback
Realized transpiration supply relative to demand
Growth and leaf responses
Free drainage
Daily outputs
LAI · biomass · grain · ET
Soil water · runoff · drainage
Soil-water carry-over
Next day → next crop / fallow


(a)
(b)
(c)
(d)
N
0
200 km
39°N
36°N
33°N
113°E
116°E
119°E
0
25
50
75
100
Conventional quota retained (%)
−4
−2
0
2
4
Targeted − uniform (t ha⁻¹ yr⁻¹)
−100
0
100
Targeted − uniform (mm yr⁻¹)
Gain
Loss
Unchanged
43.92%
37.60%
18.48%
Area-weighted median: 0
Regional mean:
+0.732 t ha⁻¹ yr⁻¹
0
10
20
30
40
50
60
Mapped rotation area (%)


(a)
Calibration (17 bins)
R² 0.64 · RMSE 0.95
Evaluation (13 bins)
R² 0.37 · RMSE 1.57
LAI (m² m⁻²)
Days after sowing
(b)
Calibration (17 bins)
R² 0.97 · RMSE 1.11
Evaluation (13 bins)
R² 0.88 · RMSE 2.58
Dry biomass (t ha⁻¹)
Days after sowing
(c)
Calibration (17 bins)
R² 0.73 · RMSE 0.86
Evaluation (16 bins)
R² 0.83 · RMSE 0.63
Daily ET (mm d⁻¹)
Days after sowing
(d)
Calibration (16 bins)
R² 0.92 · RMSE 0.52
Evaluation (16 bins)
R² 0.78 · RMSE 0.99
LAI (m² m⁻²)
Days after sowing
(e)
Calibration (15 bins)
R² 0.93 · RMSE 2.25
Evaluation (13 bins)
R² 0.91 · RMSE 2.08
Dry biomass (t ha⁻¹)
Days after sowing
(f)
Calibration (16 bins)
R² 0.74 · RMSE 0.68
Evaluation (16 bins)
R² 0.58 · RMSE 1.12
Daily ET (mm d⁻¹)
Days after sowing
Calibration observations
Evaluation observations
Calibration simulation
Evaluation simulation


(a)
Simulated seasonal ET (mm)
Observed seasonal ET (mm)
Calibration (n = 12)
R² 0.76 · RMSE 40.19
Testing (n = 4)
R² 0.91 · RMSE 73.80
(b)
Calibration (n = 12)
R² 0.62 · RMSE 37.44
Testing (n = 4)
R² 0.34 · RMSE 55.64
(c)
Simulated grain (t ha⁻¹)
Observed grain (t ha⁻¹)
Calibration (n = 12)
R² 0.82 · RMSE 1.25
Testing (n = 4)
R² 0.87 · RMSE 3.16
(d)
Calibration (n = 12)
R² 0.13 · RMSE 2.27
Testing (n = 4)
R² 0.97 · RMSE 1.70
Calibration: 2016–2018 (n = 12 per crop)
Retrospective testing: 2019 (n = 4 per crop)


(a)
Rotation dry grain (t ha⁻¹ yr⁻¹)
17
16
15
14
13
12
Uniform
Targeted
0
25
50
75
Field irrigation reduction (%)
(b)
Crop + fallow ET (mm yr⁻¹)
900
850
800
750
700
0
25
50
75
Field irrigation reduction (%)
(c)
Targeted − uniform grain (t ha⁻¹ yr⁻¹)
1.5
1.0
0.5
0.0
25% cut
50% cut
75% cut
2014
2018
2022
2025
Harvest year
(d)
Rotation dry grain (t ha⁻¹ yr⁻¹)
17
16
15
14
13
Conventional
25%
50%
75%
750
800
850
900
Crop + fallow ET (mm yr⁻¹)


(a)
Field irrigation (mm yr⁻¹)
400
350
300
250
200
2014
2018
2022
2025
Harvest year
(b)
Conventional grain retained (%)
100
98
96
94
92
90
2014
2018
2022
2025
Harvest year
(c)
ET reduction (mm yr⁻¹)
60
40
20
0
2014
2018
2022
2025
Harvest year
(d)
Conventional grain retained (%)
100
98
96
94
92
90
0
50
100
150
Field irrigation reduction (mm yr⁻¹)
95% selection target
98% selection target
Conventional quota in GRACE gaps